

Transformer-based monocular depth estimation with hybrid attention fusion and progressive regression

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ABSTRACT

Monocular depth estimation has gained prominence due to its versatile applications and minimal system requirements, despite the inherent challenges posed by its ill-posed nature. Recent advancements in Transformer techniques have significantly propelled this field forward. This paper introduces a novel Transformer-based encoder-decoder network to enhance monocular depth estimation, presenting two key improvements compared to prior approaches. Firstly, a hybrid attention fusion module is designed in the decoding stage for feature fusion. This module employs a parallel hybrid structure design to fully leverage the complementary advantages of Transformers and convolution methods, effectively capturing both local contextual and global consistency of depth maps. Additionally, to mitigate fine-grained depth loss during decoder upsampling, a deep progressive regression module is proposed. This module gradually predicts depth maps of varying resolutions by enhancing interactions between adjacent prediction branches. Experimental results on challenging indoor and outdoor benchmark datasets affirm the superior performance of the proposed method, showing that it effectively addresses some of the limitations of existing approaches.

1. Introduction

Monocular depth estimation (MDE) endeavors to ascertain the depth information between the camera and scene objects from a single RGB image. This depth information finds broad applications in autonomous driving [1], virtual reality [2], three-dimensional (3D) reconstruction [3], etc. However, MDE is fundamentally an ill-posed problem due to the absence of reliable depth cues such as motion and stereo vision relationships, rendering it a challenging subject within the realm of computer vision.

Current effective methods for MDE heavily depend on deep learning models to extract intricate information from abundant RGB images [4]. The inception of convolutional neural network (CNN)-based methods marked a significant milestone in this domain [5–16]. To enhance the accuracy of CNN-based MDE, several approaches have been explored to improve the efficacy of convolution operators, including multi-scale extraction [8], atrous convolution [6,14], channel spatial attention [9, 12,13], and feature pyramid [15]. Achieving precise dense estimation requires a model with a global receptive field for the scene alongside local detailed information. However, the inherently local nature of CNNs poses challenges in capturing long-range global information.

Recognizing the excellence of the Transformer model in modeling long-term dependencies with global receptive fields [17], researchers have introduced visual Transformer models for MDE [18–30], yielding substantial progress. Despite these accomplishments, two primary impediments hinder further advancements in Transformer-based MDE methods, serving as the motivations for our proposed approach.

Firstly, the majority of Transformer-based MDE methods [18–23] employ an encoder-decoder architecture. Encoders are typically chosen from standard image classification backbones pre-trained on ImageNet [31], and the decoder aggregates multi-scale features to generate the final depth map. While Transformers excel at modeling long-range dependencies with a global receptive field, CNNs still possess significant advantages in understanding local contextual information, particularly considering computational cost. The incorporation of advantages from both Transformers and CNNs has been demonstrated to effectively enhance model performance in various vision tasks [32–42]. However, current methods face challenges in integrating CNNs and Transformers in the decoder design, resulting in insufficient feature aggregation and a significant reduction in MDE performance.

Secondly, after the gradual upsampling of decoder features, most encoder-decoder MDE architectures only leverage the branch of the

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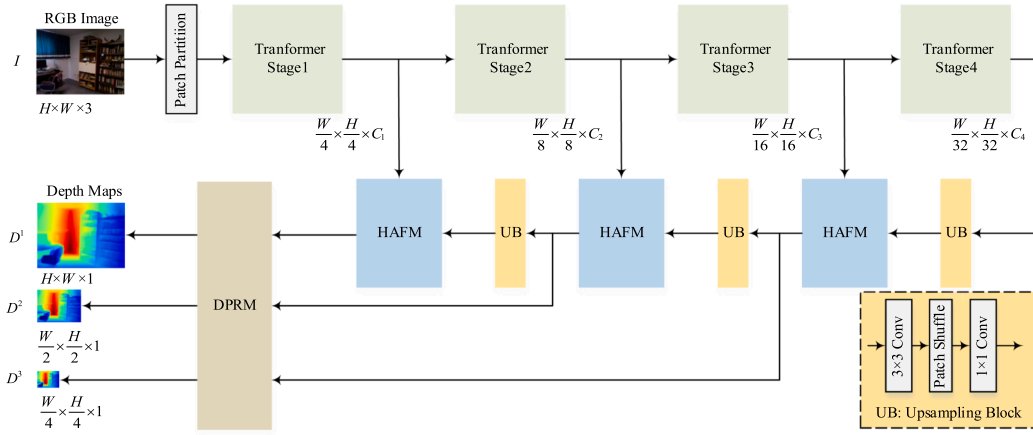


Fig. 1. Architecture of the proposed network. The network comprises an encoder and a decoder pathway, with feature fusion between the corresponding layers. Function blocks are distinguished by the rectangle with different fill colors.

highest-scale decoder feature for depth regression, leading to a loss of local depth information [24]. Gradual supervision of depth maps [43–46] from small to large can mitigate detail loss and be more conducive to predicting fine-grained depth maps. Recently, grid-style network structures [47,48] have shown great potential in fine-grained prediction due to their advantages in achieving feature interaction at different resolutions.

To address these challenges, this paper introduces a monocular depth estimation network with a Hybrid Transformer architecture. The proposed network employs the Swin Transformer [49] as the encoder, and a novel hybrid attention fusion module (HAFM) is introduced for feature fusion at each decoder stage. The HAFM leverages both Transformer and convolution to more effectively capture both local context and global consistency. Additionally, inspired by the grid-style design, we present a simple yet effective depth progressive regression module (DPRM) for estimating multi-scale depth maps using hierarchical features output by HAFM. HAFM significantly enhances feature interactions between different prediction branches through two types of skip connections, effectively preventing the attenuation of fine-grained information during the gradual upsampling process. Thanks to these design enhancements, our model demonstrates superior performance compared to state-of-the-art models.

The contributions of this paper can be summarized as follows:

- We propose a novel hybrid attention fusion module that effectively integrates the advantages of both convolution and Transformer.
- We introduce a depth progressive regression module to estimate multi-scale depth maps, mitigating loss during the gradual upsampling.
- We introduce an end-to-end Transformer-based network for monocular depth estimation, demonstrating superior performance compared to recent state-of-the-art methods on benchmark datasets for both indoor and outdoor scenes.

The remaining sections of the paper are organized as follows: [Section 2](#) reviews related work, [Section 3](#) provides a detailed explanation of the proposed method, [Section 4](#) presents the experimental results, and [Section 5](#) concludes the paper.

2. Related work

Deep learning methods have achieved significant strides in MDE, falling into two primary categories: CNN-based MDE and Transformer-based MDE. The subsequent sections offer detailed insights into each category. Additionally, a concise overview of hybrid CNN and Transformer models is presented.

2.1. CNN-based MDE methods

Eigen et al. [5] pioneered the application of CNNs to the MDE task, comprising a global coarse prediction network and a local refinement network. Subsequently, a continuum of CNN-based MDE methods have emerged [6–15]. For example, Huynh et al. [12] introduced non-local coplanarity constraints and non-local attention mechanisms to enhance the estimation performance of planar structural regions in depth maps. Both [10] and [11] employed the encoder-decoder architecture with skip connections for depth map estimation. Li et al. [14] fused different outputs from the dilated CNN in a hierarchical way to exploit the multi-scale feature for MDE. Chen et al. [15] applied a CNN framework with multi-scale feature fusion to generate high-quality depth predictions. Liu et al. [9] proposed an encoder-decoder with the combination of multi-scale fusion and joint attention mechanisms, enhancing the discriminability of feature processing. Lee et al. [8] utilized a local planar guidance operation to guide features to full scale, deviating from common upsampling layers in the decoding stage.

2.2. Transformer-based MDE methods

The emergence of Vision Transformer (ViT) architectures has demonstrated their efficiency in obtaining long-range dependencies and gained state-of-the-art performance for MDE. For example, Ranftl et al. [20] proposed a universal dense prediction Transformer model that achieved good results in semantic segmentation and monocular depth estimation. Bhat et al. [18] refined the output of CNN-based encoder-decoder backbone using a ViT with global statistical analysis. Yuan et al. [19] built a bottom-up-top-down structure with neural window CRFs as decoder and ViT serves as encoder. Kim et al. [22] utilized ViT as the encoder and designed a selective feature fusion module to fuse features in the decoder stage. Li et al. [23] and Shao et al. [25] introduced different adaptive bins strategies based on Swin Transformer backbone to boost the MDE performance. Ning et al. [24] built an encoder-decoder depth estimation network based on a novelty trap attention mechanism.

2.3. The hybrid transformer

Given that Transformer models excel in extracting global dependencies and CNNs possess significant advantages in capturing local context information, researchers strive to combine both architectures to achieve complementary benefits and enhance overall model performance. Some approaches involve embedding CNNs into the Transformer architecture. For example, Lee et al. [32] proposed a network named MPViT, which incorporates convolutional paths within the Transformer

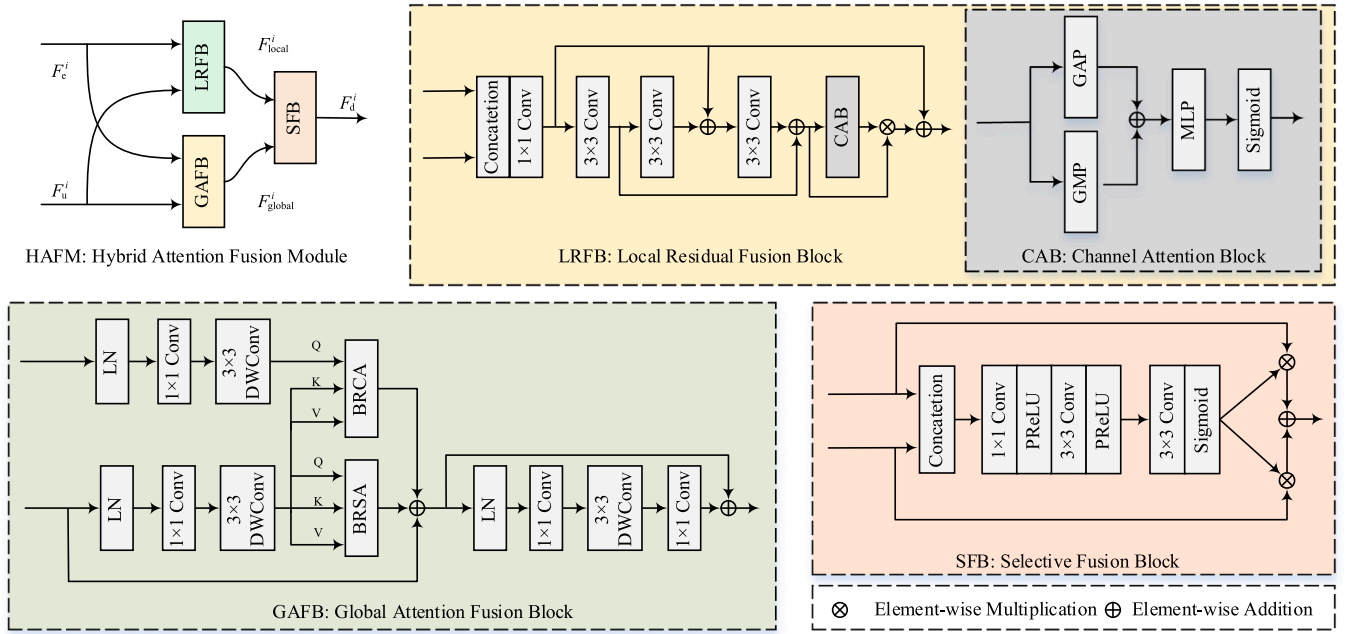


Fig. 2. Details of the proposed hybrid attention fusion module. This module includes three types of sub-blocks: the local residual fusion block, the global attention fusion block, and the selective fusion block. Transformers and convolutions are effectively used to capture local context and global consistency.

framework. Guo et al. [33] designed a novel Transformer-based hybrid network, leveraging the strengths of both Transformers and CNNs. Alternatively, some researchers adopt the Transformer and CNN as independent branches, allowing them to fully leverage the advantages of the Transformer while preventing CNNs from discarding crucial information. For instance, Li et al. [21] utilized a parallel structure of Transformer and CNN as an encoder, introducing a hierarchical aggregation and heterogeneous interaction module to combine encoder features. Yoo et al. [37] introduced an effective hybrid network for aggregating enriched features in super-resolution tasks, combining local features captured by CNNs with long-range multi-scale dependencies from Transformers.

By integrating the CNN and Transformer, the hybrid architectures are also used to leverage both local and global information for improved MDE. For example, Zhang et al. [38] proposed a joint convolutional attention and Transformer block for depth completion tasks, deeply coupling the convolutional layer and ViT into a unified block. Yang et al. [39] proposed a lightweight network that tightly couples CNN and Transformer for MDE. Bae et al. [40] examined the internal properties of representations from intermediate layers of hybrid CNN-Transformer models using synthetic datasets. Shao et al. [41] introduced uncertainty-rectified cross-distillation between CNN and Transformer models to enhance MDE.

In this paper, we follow the Swin Transformer as the encoder and improve MDE performance to some extent following two aspects bellow. Firstly, we design a parallel hybrid feature fusion in the decoding stage by combining the complementary advantages of CNN and Transformer. Secondly, we use a progressive regression strategy with efficient skip connections for depth prediction.

3. Proposed method

This section firstly presents the overall architecture of our network. A detailed description of the proposed HAFM and DPRM will be provided in subsequent sections. Finally, we specify our loss function.

3.1. Network architecture

Fig. 1 illustrates the architecture of the proposed encoder-decoder

network. In alignment with recent trends employing the Transformer for feature extraction, we utilize the Swin Transformer backbone with a four-stage pyramid structure as the encoder for processing RGB images. Given an input RGB image with resolution $H \times W$, the outputs of the four stages are the encoder features $\{F_c^1, F_c^2, F_c^3, F_c^4\}$ at scales $\{1/4, 1/8, 1/16, 1/32\}$ respectively, with channel dimensions $\{C_1, C_2, C_3, C_4\} = \{C, 2C, 4C, 8C\}$, where C is set to 192 as per the configuration in [49]. In the decoding stage, the upsampling block (UB) depicted in Fig. 1 is used between each two adjacent stages to upsample the lower-resolution feature map for feature fusion. The UB comprises a 3×3 convolution for feature channel reduction, along with a combination of a patch shuffle layer and a 1×1 convolution for feature upsampling.

Three proposed HAFMs fuse the upsampling features with the corresponding scale encoder features. The three sets of decoder features generated by the HAFMs are input to the DPRM. The DPRM progressively predicts three sets of depth maps $\{D^3, D^2, D^1\}$ with resolutions $\{H/4 \times W/4, H/2 \times W/2, H \times W\}$, respectively.

3.2. Hybrid attention fusion module

To efficiently integrate the strengths of both CNN and Transformer during the hierarchical upsampling of the decoder, we have devised a HAFM, as depicted in Fig. 2. The HAFM adopts a hybrid Transformer design. It consists of three types of sub-blocks: the local residual fusion block (LRFB), the global attention fusion block (GAFB), and the selective fusion block (SFB), as also illustrated in Fig. 2. The encoder feature and the upsampling feature are directed to the LRFB and GAFB, respectively. The GAFB is structured based on a Transformer design, ensuring the effective fusion of global contextual information, while the LRFB is designed based on a CNN structure to ensure effective fusion of local details. Subsequently, the output of the

GAFB and LRFB are fed into the SFB to achieve adaptive fusion of the global feature and local feature.

As shown in Fig. 2, the GAFB leverages multi-head self-attention (MHSA) and cross-attention mechanisms to process the input features. The GAFB initially normalizes encoder feature F_c^i and upsampling feature F_u^i separately. Subsequently, a 1×1 convolution and a 3×3 depth-separable convolution encodes relative position information more

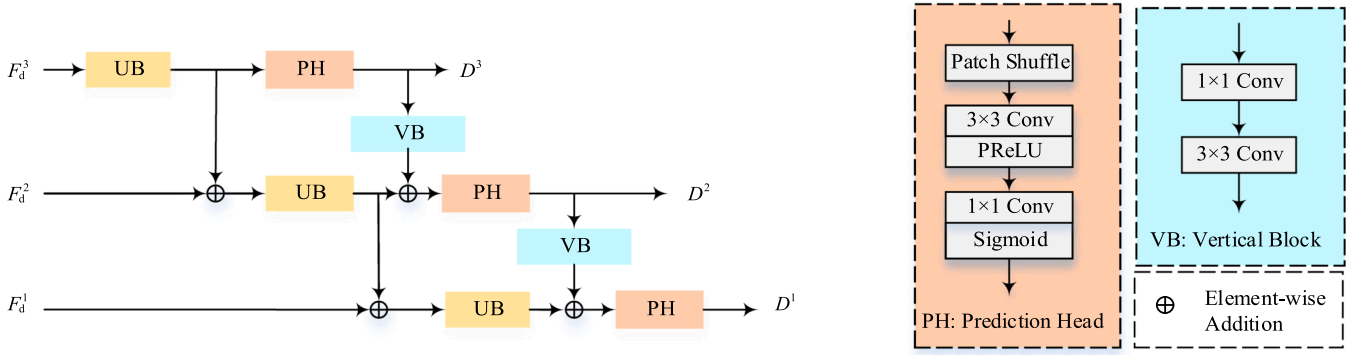


Fig. 3. Details of the proposed depth progressive regression module. This module establishes three backbone regression branches and implements two types of skip connections between adjacent prediction branches.

effectively, facilitating subsequent global attention calculations [50]. To mitigate computational complexity, GAFB employs the global attention calculation method proposed by BiFormer [51] for the bi-level routing self-attention (BRSA) of the upsampling feature and the bi-level routing cross-attention (BRCA) between the encoder feature and upsampling feature. BiFormer achieves content-aware sparse patterns through region-level routing and token-level attention steps in a query-adaptive manner, resulting in a superior computational complexity of $O((H \times W)^{4/3})$, while the commonly used multi-head self-attention has a computational complexity of $O((H \times W)^2)$. The sum of the two global attentions is input into a feed forward network with a residual connection, completing the generation of global feature F_{global}^i . The fusion of cross-attention and self-attention within GAFB comprehensively integrates features from a global perspective, enhancing its global representation capability. The procedure is outlined as follows:

$$F_{t1}^i = BRCA(W_{\text{en}}(LN(F_e^i)), W_{\text{en}}(LN(F_u^i))) + BRSA(W_{\text{en}}(LN(F_u^i))) + F_u^i \quad (1)$$

$$F_{\text{global}}^i = F_{t1}^i + FFN(LN(F_{t1}^i)) \quad (2)$$

where $LN(\cdot)$ denotes the normalization layer. $FFN(\cdot)$ denotes the feed forward network. F_{t1}^i denotes the temporary feature. $W_{\text{en}}(\cdot)$ denotes two serially arranged convolutional layers for encoding purpose, comprising a 1×1 convolution and a 3×3 depth separable convolution. Both $BRSA(\cdot)$ and $BRCA(\cdot)$ calculate the global attention using the method provided by BiFormer [51].

The LRFB uses a composite residual structure and a convolutional channel attention mechanism for feature fusion. First, features are aggregated through concatenation and a 1×1 convolution. Then, three sets of 3×3 convolutions within the composite residual structure enhance local information. A standard channel attention block (CAB) [52] is integrated to adaptively rescale the channel features. The CAB aggregates spatial information through global average-pooling (GAP) and global max-pooling (GMP) operations, followed by the utilization of a multi-layer perceptron (MLP) and Sigmoid operation to generate the channel attention map. Finally, LRFB effectively processes local regions and generates the local feature F_{local}^i during feature fusion using the aforementioned traditional CNN operations. The process of feature fusion in LRFB is outlined as follows:

$$F_{t2}^i = W_{1 \times 1}(\text{concat}(F_u^i, F_e^i)) \quad (3)$$

$$F_{\text{local}}^i = F_{t2}^i + W_{\text{Res}}(F_{t2}^i) \cdot CA(W_{\text{Res}}(F_{t2}^i)) \quad (4)$$

where $\text{concat}(\cdot)$ denotes the concatenation. $CA(\cdot)$ denotes the channel attention calculation.

The SFB adaptively fuses local and global features extracted from the

LRFB and GAFB by generating spatial attention maps. These maps exploit interdependencies between local and global features, optimizing the fusion process. Initially, the two input features are concatenated along the channel dimension and subjected to a series of convolution operations for feature extraction. Subsequently, a 3×3 convolution and a Sigmoid function are applied to produce two channel spatial attention maps (SA_i). Finally, we multiply the two attention maps with the corresponding input features and add the results to generate decoder feature. The procedural steps are outlined as follows:

$$SA_1, SA_2 = \delta(W_{3 \times 3}(W_{\text{ext}}(\text{concat}(F_{\text{local}}^i, F_{\text{global}}^i)))) \quad (5)$$

$$F_d^i = SA_1 \cdot F_{\text{local}}^i + SA_2 \cdot F_{\text{global}}^i \quad (6)$$

where $\delta(\cdot)$ denotes the Sigmoid function.

3.3. Depth progressive regression module

In contrast to conventional deep regression methods that exclusively leverage the branch of the highest-scale feature for regression purposes, our proposed DPRM adopts a novel approach by predicting three sets of depth maps with varying resolutions, progressively ranging from small to large. This progressive depth regression strategy proves advantageous in mitigating the loss associated with detailed depth information, ultimately yielding more refined depth maps.

As depicted in Fig. 3, the structure of the DPRM is inspired by the grid-style design [47,48]. To establish three backbone regression branches, we employ a series connection of an UB and a prediction head (PH). Subsequently, two types of skip connections are introduced between adjacent prediction branches: the feature-to-feature skip connection and the depth-to-feature skip connection. The former links the output of each UB to the same-scale feature of the adjacent regression backbone, enhancing the expression of each scale feature. The latter, known as the depth-to-feature skip connection, establishes a connection between the depth map and the input of the PH with an adjacent scale. This connection is achieved by applying the vertical block (VB) shown in Fig. 3 to the depth map. The feature-to-feature skip connection enriches the expression of each scale feature, while the depth-to-feature skip connection renders the input feature of each PH more meaningful. The processing of features in the DPRM can be described as:

$$D^3 = PH(UB(F_d^3)) \quad (7)$$

$$F_{t3}^2 = UB(F_d^3) + F_d^2 \quad (8)$$

$$D^2 = PH(UB(F_{t3}^2) + VB(D^3)) \quad (9)$$

$$F_{t4}^1 = UB(F_d^2) + F_d^1 \quad (10)$$

Table 1

Performance evaluation on NYU-V2. The best scores are highlighted in bold font.

Methods	Error (lower is better)			Accuracy (higher is better)		
	Abs Rel	RMSE	Log10	δ_1	δ_2	δ_3
Fu et al. [6]	0.114	0.510	0.052	0.829	0.966	0.992
Yin et al. [7]	0.108	0.416	0.048	0.875	0.976	0.994
Lee et al. [8]	0.110	0.392	0.047	0.885	0.978	0.994
Patil et al. [50]	0.104	0.356	0.038	0.898	0.981	0.996
Xiao et al. [16]	0.104	0.380	0.045	0.895	0.983	0.996
Bhat et al. [18]	0.103	0.364	0.044	0.903	0.984	0.997
Ranftl et al. [20]	0.110	0.357	0.045	0.904	0.988	0.998
Yang et al. [28]	0.106	0.365	0.045	0.900	0.983	0.996
Li et al. [30]	0.106	0.349	-	0.905	0.989	0.998
Yuan et al. [19]	0.095	0.334	0.041	0.922	0.992	0.998
Li et al. [21]	0.096	0.339	0.041	0.921	0.989	0.998
Li et al. [23]	0.094	0.330	0.040	0.925	0.989	0.997
Ning et al. [24]	0.090	0.329	0.039	0.927	0.991	0.998
Agarwal et al. [26]	0.090	0.322	0.039	0.929	0.991	0.998
Shao et al. [27]	0.087	0.311	0.038	0.936	0.991	0.998
Ours	0.086	0.316	0.036	0.936	0.992	0.998

$$L_{\text{depth}} = \alpha \sqrt{\frac{1}{n} \sum_i (\lg y_i - \lg y_i^*)^2 - \frac{\lambda}{n^2} \left(\sum_i (\lg y_i - \lg y_i^*) \right)^2} \quad (12)$$

where n denotes the number of ground truth pixels in a depth map. Following [21], λ and α are set to be 0.85 and 10, respectively.

2) Normal loss. To compute the mean plane loss and capture more structural prediction differences, we evaluate the surface normal difference for all non-overlapping patches. The calculation of the normal loss is as follows:

$$L_{\text{normal}} = \frac{1}{n} \sum_i \|N_i - N_i^*\|_2^2 \quad (13)$$

where N_i and N_i^* are the surface normals of the predicted depth map and the corresponding ground truth. Surface normals are computed by stacking the 3D depth points within the patch and solving a closed-form solution for a least-squares problem, as outlined in [53,54].

4. Experimental results

In this section, we conducted experimental comparisons to evaluate the performance of the proposed method in comparison to state-of-the-art techniques. Additionally, we analyzed the impact of the proposed HAFM and DPRM through ablation studies.

4.1. Experimental setup

Datasets. We chose the NYU-V2 [55] dataset for indoor scenes and the KITTI dataset [56] for outdoor scenes for both training and testing. To assess the generalization of the generated indoor model, we utilized the iBims-1 [57] dataset specifically designed for indoor scenes.

We employed the official split method [5] to split NYU-V2 and KITTI datasets. Specifically, for NYU-V2, approximately 50,000 image sets from 249 scenes were selected for training, while 654 image sets from

$$D^1 = PH(UB(F_{14}^1) + VB(D^2)) \quad (11)$$

where $UB(\cdot)$ denotes the upsampling block. $PH(\cdot)$ denotes the prediction head. $VB(\cdot)$ denotes the vertical block.

3.4. Loss function

The loss function plays a crucial role in adjusting network parameters during training, exerting a substantial influence on the ultimate estimation accuracy. In our model, the loss function comprises the sum of a depth loss and a normal loss. Each loss term is elucidated below.

1) Depth loss. Following previous Transformer-based MDE methods [18,21], we use the scale invariant logarithmic loss to supervise our network. Given the ground truth depth y_i^* and the predicted depth y_i at a pixel location i . The depth loss function is calculated as:

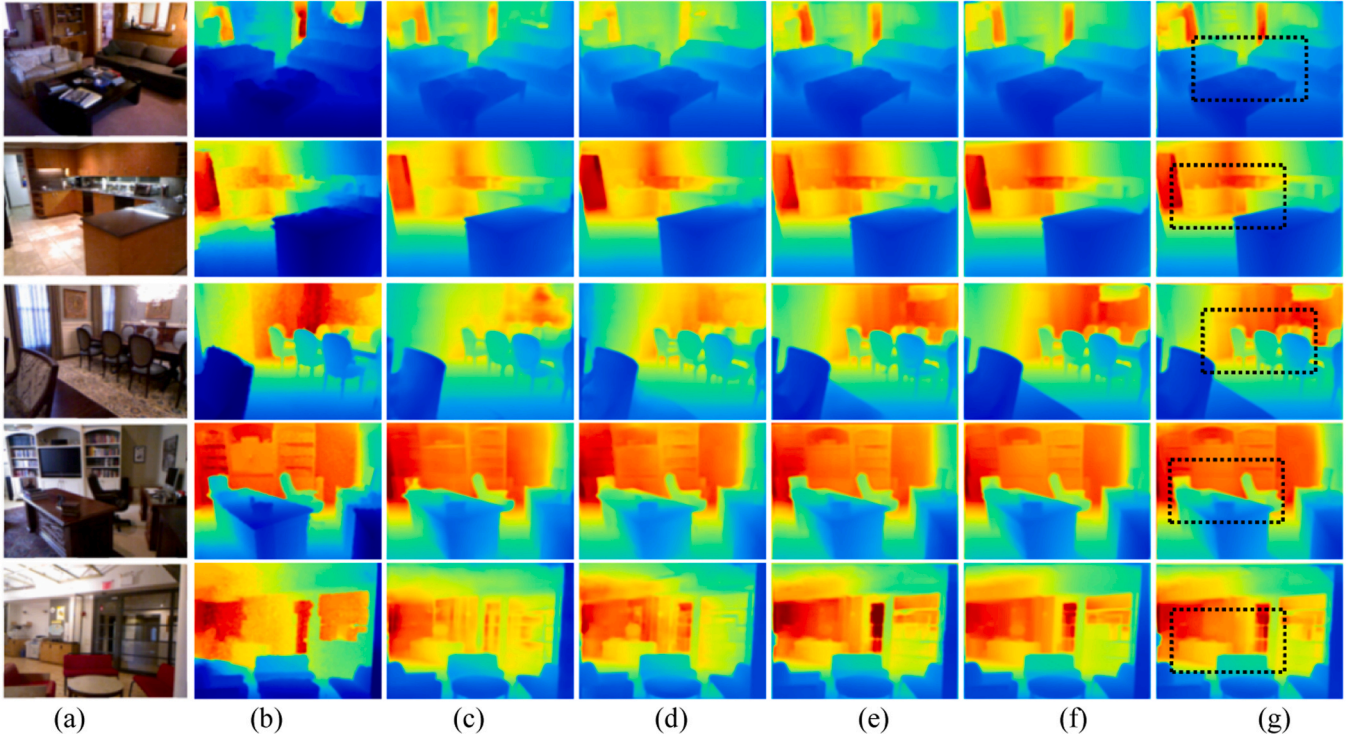


Fig. 4. Qualitative comparison results on NYU-V2. Color indicates depth (red is far, blue is close). (a) RGB images, (b) ground truth depth maps, results of (c) Lee et al. [8], (d) Bhat et al. [18], (e) Yuan et al. [19], (f) Shao et al. [27], and (g) the proposed method.

Table 2

Performance evaluation on KITTI. Depths are capped at 80 m. The best scores are highlighted in bold font.

Methods	Error (lower is better)			Accuracy (higher is better)		
	Abs Rel	RMSE	Sq Rel	δ_1	δ_2	δ_3
Fu et al. [6]	0.071	2.728	0.306	0.932	0.985	0.994
Yin et al. [7]	0.072	3.258	0.302	0.938	0.990	0.998
Lee et al. [8]	0.060	2.798	0.249	0.955	0.993	0.998
Patil et al. [50]	0.071	2.842	0.270	0.953	0.993	0.998
Xiao et al. [16]	0.057	2.415	0.201	0.964	0.996	0.999
Bhat et al. [18]	0.058	2.360	0.190	0.964	0.995	0.999
Ranftl et al. [20]	0.062	2.573	0.092	0.959	0.995	0.999
Yang et al. [28]	0.064	2.755	0.252	0.956	0.994	0.999
Li et al. [30]	0.060	2.325	-	0.964	0.996	0.999
Yuan et al. [19]	0.052	2.129	0.155	0.974	0.997	0.999
Li et al. [21]	0.052	2.143	0.158	0.975	0.997	0.999
Li et al. [23]	0.052	2.098	0.151	0.974	0.997	0.999
Ning et al. [24]	0.050	1.869	0.128	0.980	0.998	0.999
Agarwal et al. [26]	0.051	2.081	0.149	0.976	0.997	0.999
Shao et al. [27]	0.050	2.025	0.141	0.978	0.998	0.999
Ours	0.050	2.011	0.127	0.981	0.998	0.999

215 scenes were used for testing. Both depth maps and RGB images were centered and cropped into 608×456 pixels to eliminate edge areas with significant depth deviation. For KITTI, 22,000 images sets from 28 scenes were utilized for training, while 697 images sets from different scenes were reserved for testing. Both depth maps and RGB images were resized to 1216×352 pixels. Loss evaluation was conducted solely on valid LIDAR points of depth maps and we capped the maximum predictions of all networks to a maximum depth of 80 m. The training dataset enhancement operations for NYU-V2 and KITTI followed the approach referenced in [8].

The iBims-1 dataset comprises 100 high-quality RGB-D images. In comparison to NYU-V2, iBims-1 exhibits a similar depth range, features distinct indoor scenes, and boasts higher-quality depth maps.

Implementation Details. Our network implementation was carried out on Ubuntu 18.04, utilizing the PyTorch framework. Training took place on two NVIDIA RTX A4000 GPUs with a batch size of 8. For

training optimization, we employed the Adam optimizer to adjust network parameters. The initial learning rate was set at 0.0001, reduced by 10 % every 5 cycles, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a decay rate of 0.0001. The training epoch was configured for 30 cycles.

Evaluation Metrics. We employed standard error and accuracy evaluation metrics commonly utilized in MDE studies for quantitative assessment. Given the predicted depth y_i , and the ground truth depth y_i^* at a pixel i , and n denoting the total number of pixels in an image, the

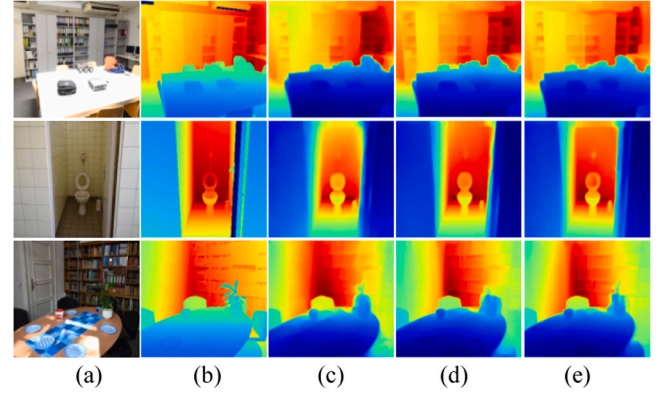


Fig. 6. Qualitative comparison results on iBims-1. Color indicates depth (red is far, blue is close). (a) RGB images, (b) ground truth depth maps, results of (c) Lee et al. [8], (d) Shao et al. [27], and (e) the proposed method.

Table 3

Performance evaluation on iBims-1. The best scores are highlighted in bold font.

Methods	Error (lower is better)			Accuracy (higher is better)		
	Abs Rel	RMSE	Log10	δ_1	δ_2	δ_3
Lee et al. [8]	0.288	2.452	0.178	0.181	0.568	0.879
Shao et al. [27]	0.262	2.652	0.162	0.195	0.623	0.884
Ours	0.238	2.442	0.159	0.231	0.642	0.888



Fig. 5. Qualitative comparison results on KITTI. Color indicates depth (yellow is far, purple is close). (a) RGB images, (b) ground truth depth maps, results of (c) Lee et al. [8], (d) Bhat et al. [18], (e) Shao et al. [27], and (f) the proposed method.

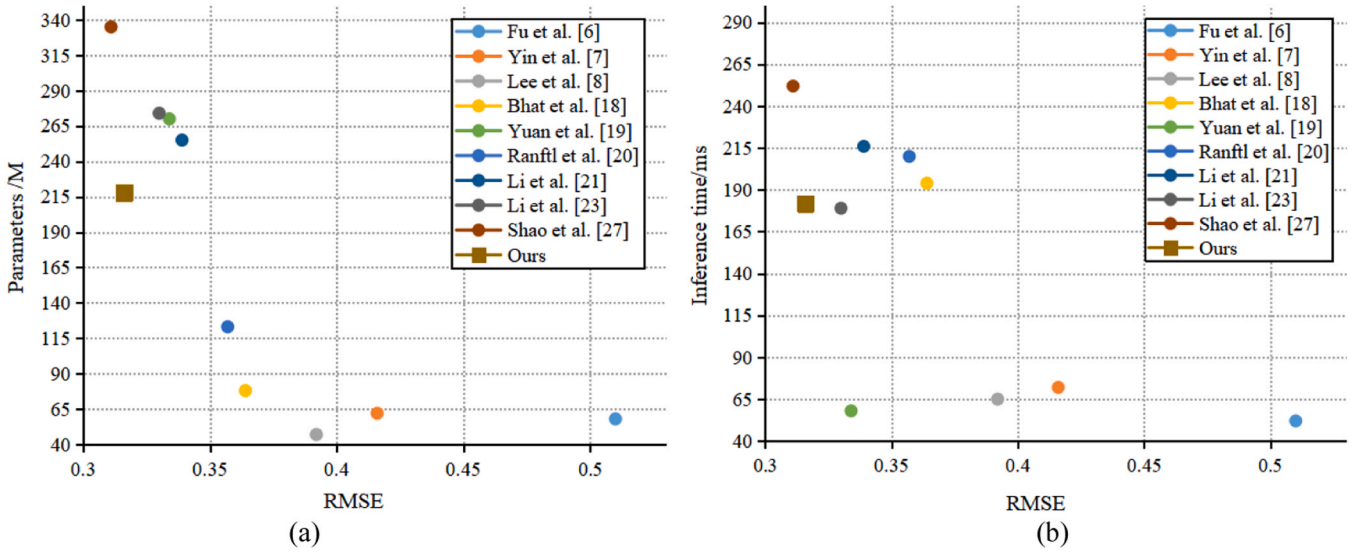


Fig. 7. Model efficiency comparisons. (a) Model parameter vs. RMSE. (b) Inference time vs. RMSE.

Table 4

Quantitative comparison results of different feature fusion methods. w/o means without.

Methods	Error (lower is better)			Accuracy (higher is better)		
	Abs Rel	RMSE	Log10	δ_1	δ_2	δ_3
LRFB	0.117	0.541	0.051	0.862	0.971	0.992
LRFB w/o CAB	0.118	0.544	0.053	0.860	0.969	0.992
GAFB	0.114	0.527	0.049	0.870	0.974	0.992
LRFB with MHSA	0.115	0.529	0.050	0.868	0.973	0.992
SFB	0.117	0.525	0.050	0.862	0.973	0.993
SFB w/o SA	0.117	0.528	0.051	0.860	0.971	0.992
HAFM	0.086	0.316	0.036	0.936	0.992	0.998

error evaluation metrics are defined as follows.

Abs Relative difference:

$$\text{AbsRel} = \frac{1}{n} \sum_i \frac{|y_i - y_i^*|}{y_i^*} \quad (14)$$

Squared Relative difference:

$$\text{SqRel} = \frac{1}{n} \sum_i \frac{\|y_i - y_i^*\|^2}{y_i^*} \quad (15)$$

Mean Log10 error:

$$\text{Log10} = \frac{1}{n} \sum_i |\log_{10} y_i - \log_{10} y_i^*| \quad (16)$$

Root Mean Squared Error:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - y_i^*)^2} \quad (17)$$

The accuracy evaluation metrics (Threshold Accuracy) is defined as follows:

$$\text{TA} = \frac{1}{n} \sum_i g(y_i, y_i^*) \quad (18)$$

where

$$g(y_i, y_i^*) = \begin{cases} 1, & \delta = \max\left(\frac{y_i^*}{y_i}, \frac{y_i}{y_i^*}\right) < \text{thr} \\ 0, & \text{otherwise} \end{cases} \quad (19)$$

TA describes the ratio of maximum relative error δ below threshold thr . Conditions $\delta < 1.25$, $\delta < 1.25^2$, and $\delta < 1.25^3$ were used in the experiment, denoted as δ_1 , δ_2 , δ_3 , respectively.

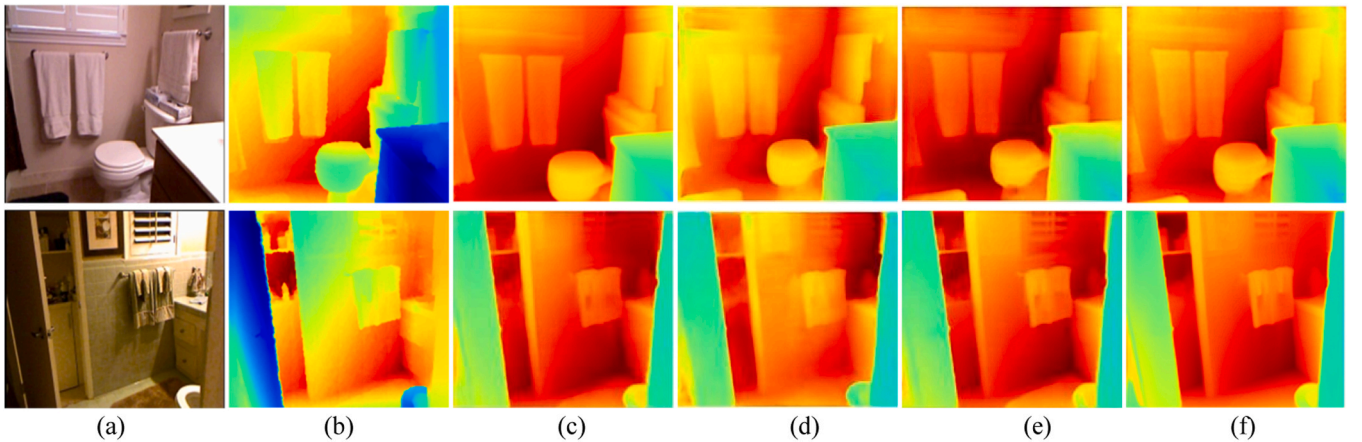


Fig. 8. Qualitative comparison results of different feature fusion methods. Color indicates depth (red is far, blue is close). (a) RGB images, (b) ground truth depth maps, results of (c) LRFB, (d) GAFB, (e) SFB, and (f) HAFM.

Table 5

Quantitative comparison results of different depth regression methods. F-F means feature-to-feature and D-F means depth-to-feature.

Methods	Error (lower is better)			Accuracy (higher is better)		
	Abs Rel	RMSE	log10	δ_1	δ_2	δ_3
One branch	0.114	0.534	0.045	0.912	0.981	0.995
Without skip connection	0.111	0.527	0.049	0.917	0.986	0.995
Only F-F skip connection	0.089	0.318	0.040	0.934	0.990	0.998
Only D-F skip connection	0.090	0.316	0.039	0.932	0.991	0.998
DPRM	0.086	0.316	0.036	0.936	0.992	0.998

4.2. Performance comparison

4.2.1. Experiment on NYU-V2

Table 1 presents results of the evaluation metrics, model parameters, and inference time. The results of the compared methods were sourced from the literature. The initial four rows correspond to CNN-based methods, while the subsequent rows are all Transformer-based methods. The table highlights that our method achieved optimal performance in accuracy metrics and the best performance in two error metrics. Specifically, our method improves Abs Rel by 1.2 % and Log10 by 5.3 % compared to the suboptimal result, while RMSE is 1.6 % higher than the optimal result.

Fig. 4 presents qualitative comparison results, illustrating the superior predictive performance of our method, particularly in capturing local geometric details such as depth boundaries of tables, chairs, and home furnishings. Moreover, our method excels in capturing essential global features of depth maps, encompassing overall depth distribution and scene depth variations. This dual capability to predict both local and global scene depth information is evident in both qualitative and quantitative assessments, affirming the enhanced accuracy of indoor depth estimation achieved by the proposed method.

4.2.2. Experiment on KITTI

Our method underwent evaluation on the KITTI dataset, with the corresponding metrics detailed in Table 2. Remarkably, our method surpasses the majority of competitors, achieving state-of-the-art results in five

evaluation metrics. The only exception is the RMSE metric, which is marginally inferior to that reported by Ning et al. [24]. Qualitative comparison results are presented in Fig. 5, demonstrating the commendable performance of our method in capturing fine-grained depth information, including pedestrians and vehicles, as well as

depicting global depth changes within the scene.

4.2.3. Cross-dataset validation

To demonstrate the generalization capability of the proposed network, we evaluated the generated indoor model using the iBims-1 dataset. Among the selected comparison methods, Lee et al. [8] is the CNN-based methods, and Shao et al. [27] is the second best among the Transformer-based methods after the proposed method. It is noteworthy that the comparison methods were trained and generated on the NYU-V2 dataset without any parameter adjustments.

Fig. 6 and Table 3 presents both qualitative and quantitative comparison results. Our method surpasses the competitors across all performance metrics and achieves notably superior visual results. Consequently,

our method exhibits enhanced generalization.

4.2.4. Model efficiency comparison

Inference time and model parameters are crucial metrics for evaluating the efficiency of MDE methods. We

conducted a trade-off analysis between RMSE, model parameters, and inference time, as visualized in Fig. 7. The results of the compared methods were obtained by executing their open-source codes on our test machine. From Fig. 7, it is evident that, compared to other Transformer-based methods, our approach achieves a superior balance in depth

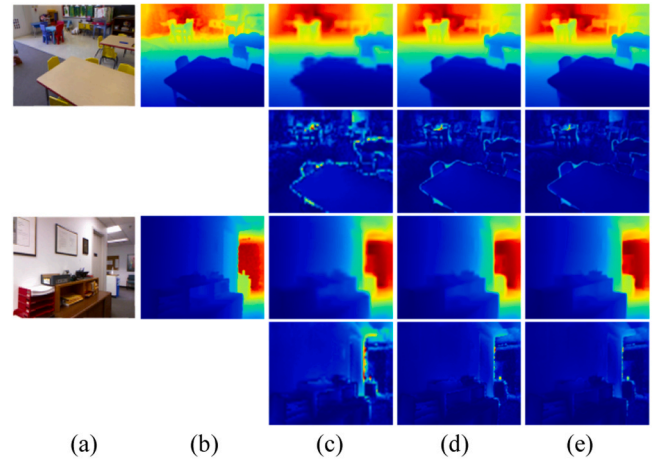


Fig. 10. Qualitative comparison of different resolution outputs by DPRM. Color indicates depth or value (red is far or big, blue is close or small). (a) RGB images, (b) ground truth depth maps, predicted depth maps and corresponding absolute prediction errors for the branches of (c) D^3 , (d) D^2 , and (e) D^1 .

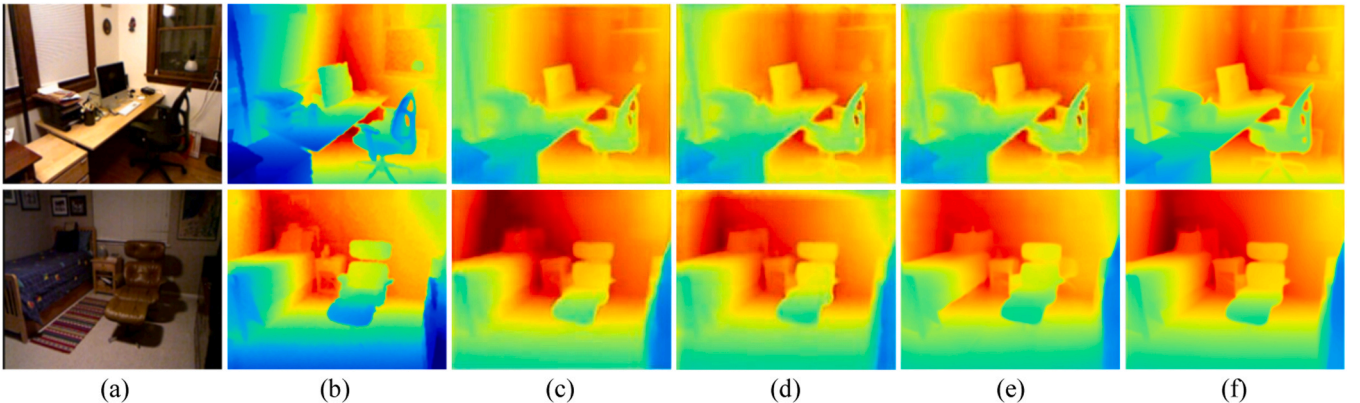


Fig. 9. Qualitative comparison results of different depth regression methods. Color indicates depth (red is far, blue is close). (a) RGB images, (b) ground truth depth maps, results of (c) one branch, (d) without skip connection, (e) only D-F skip connection, and (f) DPRM.

estimation performance metrics, model parameters, and inference time. While the RMSE of Shao et al. [27] is only 0.05 better than ours, its inference time is approximately 70 ms longer, and its model parameters exceed ours by about 120 million. Although our method carries a disadvantage in terms of parameters and inference time compared with CNN-based methods, quantitative evaluation metrics of our method outperform those of the CNN-based methods by a considerable margin.

4.3. Ablation study

Several ablation experiments were undertaken to assess the effectiveness and rationale behind the proposed HAFM and CPRM. All experiments were executed on the NYU-V2 dataset.

4.3.1. Effectiveness of HAFM

Table 4 presents the quantitative comparison results for different feature fusion methods in the decoder stage. Evidently, the internal attention mechanisms used by LRFB, GAFB, and SFB make each block more efficient at processing features, thereby improving predictive performance. The HAFM combines LRFB, GAFB, and SFB in a rational manner, resulting in superior depth estimation performance compared to using any of them individually. Compared to the suboptimal result, HAFM improves Abs Rel by 24.5 %, Log10 by 68.4 %, and RMSE by 26.5 %.

Visual results in Fig. 8 demonstrate the efficacy of our HAFM in comparison to alternative feature fusion methods. HAFM exhibits a more comprehensive and superior depth estimation performance. Specifically, when contrasted with GAFB, HAFM excels in capturing finer details of object boundaries. Additionally, in comparison to LRFB, HAFM outperforms in predicting global depth changes.

4.3.2. Effectiveness of DPRM

Table 5 presents quantitative comparisons of various depth regression methods within the network. The term “one branch” denotes utilizing the branch with the highest-scale fused feature for depth regression, while “without skip connections” indicates using three backbone prediction branches without any skip connection for depth regression. The table illustrates that employing only the highest-scale feature branch for depth estimation performs the least favorably among the compared methods. Adding skip connections between adjacent prediction branches enhances feature interactions and therefore significantly improves performance, and the skip connections used in our DPRM achieve optimal results. In Fig. 9, visual comparisons of various depth regression methods are presented. Since the performance indicators of feature-to-feature skip connection and depth-to-feature skip connection are nearly equal, only the depth-to-feature skip connection was chosen for qualitative analysis. The visual comparison results demonstrate the superior performance of the proposed DPRM, especially in estimating depth maps with finer-grained structural details of the furniture.

Fig. 10 shows a visual comparison of three different resolution depth maps (D^1 , D^2 , and D^3 in Fig. 3) output by DPRM. For convenience, we adjusted the resolution of each image to the same size. It is evident that the absolute prediction errors for deep edges in D^1 is lower than in D^2 and D^3 , indicating that DPRM effectively mitigates the prediction loss of fine-grained depth information during the decoder’s layer-by-layer upsampling process.

5. Conclusion and future work

This paper introduces a Transformer-based MDE network incorporating hybrid attention fusion and depth progressive regression. The network includes a HAFM based on a hybrid Transformer design. The HAFM, rooted in a parallel hybrid Transformer and CNN design, effectively synergizes convolutional and Transformer strengths, elevating feature fusion within the decoder stage. Additionally, a DPRM is devised

to progressively regress depth maps at different scales. Experimental results on NYU-V2 and KITTI datasets demonstrated the superior performance of our proposed methods. A series of ablation experiments validated the efficacy and rationale of the introduced HAFM and DPRM. Future research endeavors will explore the integration of the proposed network into a monocular visual simultaneous localization and mapping system for both indoor and outdoor scenes.

CRedit authorship contribution statement

Nan Gao: Validation, Supervision, Software. **Zhaozong Meng:** Supervision, Methodology, Formal analysis, Conceptualization. **Zonghua Zhang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Peng Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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